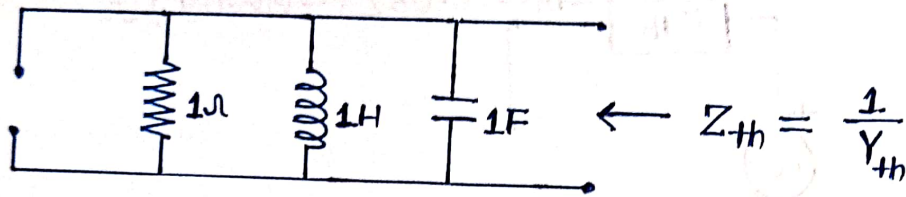


CTD QUIZ-1 (Set A)

Solution

Q1(a): Thevenin's equivalent circuit seen from the terminal of the rightmost 1Ω resistor ($\omega = 30 \text{ rad/sec}$)

Step(I): Turn off/open it, the independent current source to get Thevenin impedance (Z_{th}), we get-



(1M) $\left\{ \begin{aligned} \text{Here, } Y_R &= \frac{1}{R} = \frac{1}{1} = 1 \text{ S} \\ Y_L &= \frac{-j}{X_L} = \frac{-j}{\omega L} = \frac{-j}{30 \times 1} = \frac{-j}{30} \text{ S} \\ Y_C &= \frac{j}{X_C} = j\omega C = j30 \times 1 = j30 \text{ S} \end{aligned} \right.$

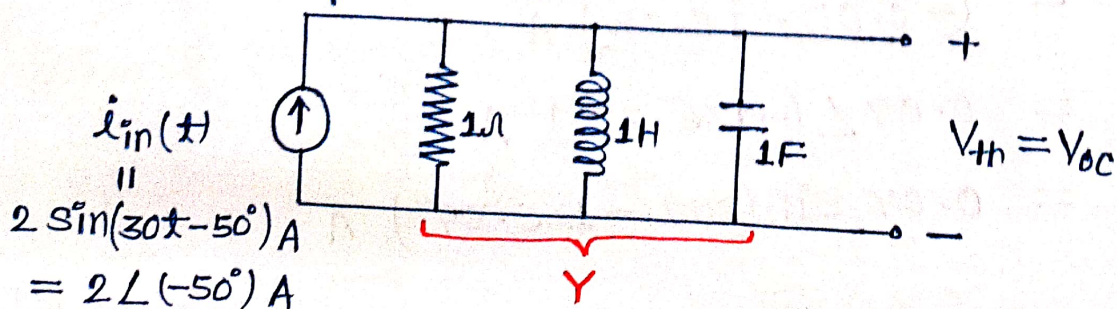
$\therefore Y_{th} = Y_R + Y_L + Y_C = 1 - \frac{j}{30} + j30 = (1 + j29.97) \text{ S}$

Thevenin Impedance, $Z_{th} = \frac{1}{Y_{th}} = \frac{1}{(1 + j29.97)}$

$= (0.00111 - j0.03333) \Omega$

$= 0.03 \angle (-88.09^\circ) \Omega \rightarrow (2M)$

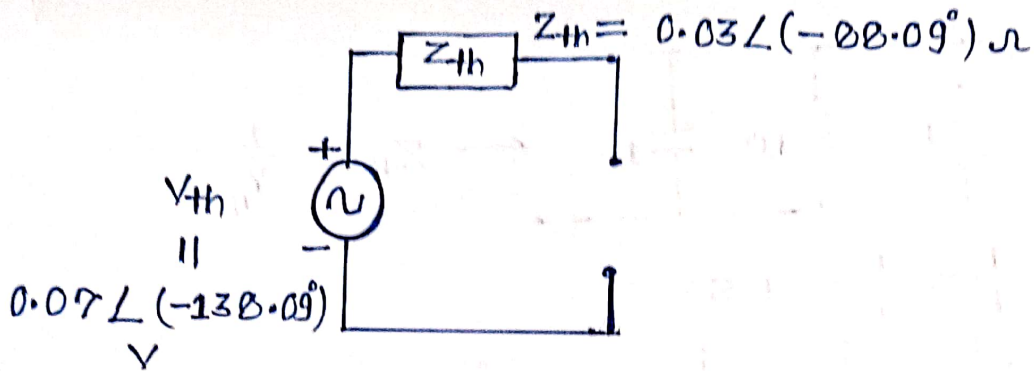
Step(II): To get Thevenin voltage (V_{th}) / open-circuit node voltage, we get-



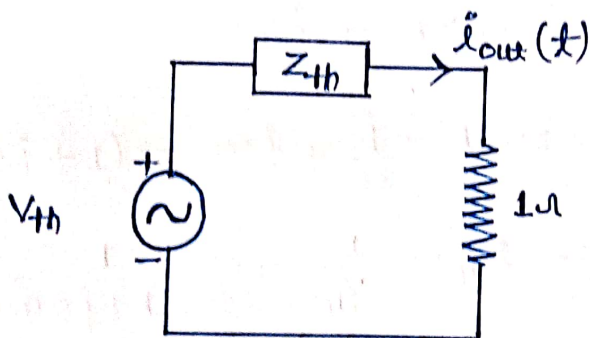
$$\begin{aligned}
 V_{th} &= \frac{I_{in}}{Y} = \frac{I_{in}}{Y_{th}} = \frac{2 \angle (-50^\circ)}{(1 + j29.97)} = \frac{2 \angle (-50^\circ)}{29.99 \angle 88.09^\circ} \\
 &= 0.07 \angle (-138.09^\circ) \text{ V} \\
 &= (-0.05 - j0.04) \text{ V} = 0.07 \sin(30t - 138.09^\circ) \text{ V}
 \end{aligned}$$

+ (1) → (4M)

Now, Thevenin's equivalent circuit seen from the terminals of the rightmost 1Ω resistor;



Q1 (b) :-



$$\begin{aligned}
 \therefore i_{out}(t) &= \frac{V_{th}}{Z_{th} + 1} \\
 &= \frac{0.07 \angle (-138.09^\circ)}{0.03 \angle (-88.09^\circ) + 1} \\
 &= (-0.05 - j0.05) \text{ A} \\
 &= 0.07 \angle (-136.37^\circ) \text{ A} \\
 &= 0.07 \sin(30t - 136.37^\circ) \text{ A}
 \end{aligned}$$

+ (1.5M)

$$\therefore v_c(t) = i_{out}(t) \times 1\Omega$$

$$= 0.07 \sin(30t - 136.37^\circ) \text{ V} \quad + \textcircled{1.5}$$

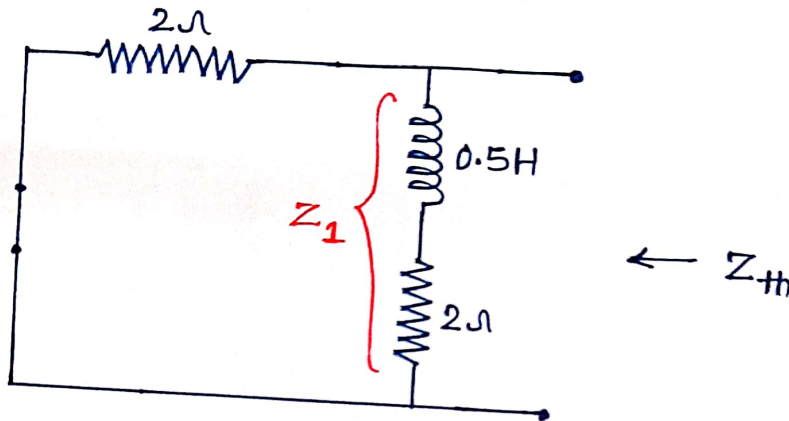
→ (1.5M)

CTD QUIZ-1 (Set B)

Solution

Q1(a) ÷ $\omega = 1 \text{ rad/sec}$

Step(I): To get Thevenin impedance (Z_{th}), short circuit the independent voltage source, we get -

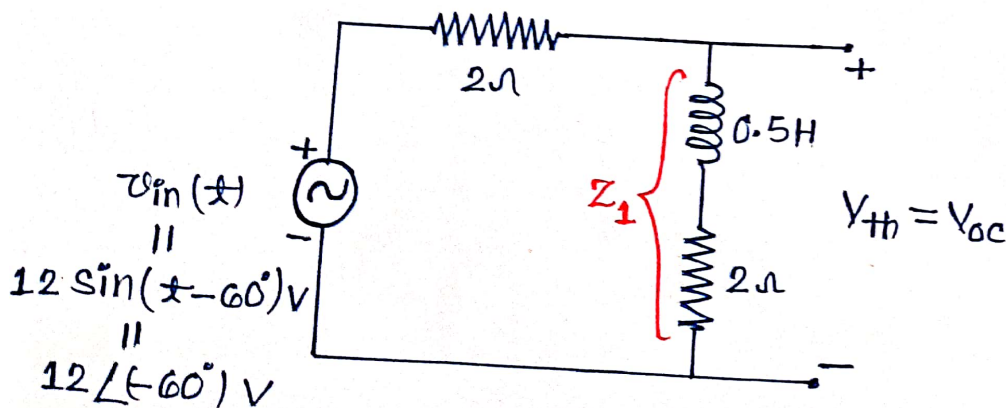


$$Z_1 = 2 + jX_L = 2 + j\omega L = 2 + j \times 1 \times 0.5 = (2 + j0.5)\Omega \quad \text{--- (1)}$$

$$\therefore Z_{th} = \frac{2Z_1}{2 + Z_1} = \frac{2(2 + j0.5)}{(2 + 2 + j0.5)} = (1.02 + j0.12)\Omega \quad \text{--- (2)}$$

$$= 1.02 \angle 6.91^\circ \Omega \quad \text{--- (2)}$$

Step(II): To get thevenin voltage (V_{th}) / open circuit node voltage across terminals of the capacitor, we get -



$$\therefore V_{th} = \left(\frac{Z_1}{Z_1 + 2} \right) v_{in}(t)$$

$$= \frac{(2 + j0.5)}{(2 + j0.5 + 2)} 12 \angle (-60^\circ)$$

$$= (0.51 + j0.06) \times 12 \angle (-60^\circ)$$

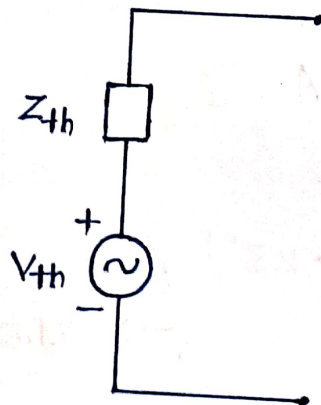
$$= 0.51 \angle 6.91^\circ \times 12 \angle (-60^\circ)$$

$$= (3.69 - j4.91) \text{ V}$$

$$= 6.14 \angle (-53.09^\circ) \text{ V}$$

$$= 6.14 \sin(t - 53.09^\circ) \text{ V} \rightarrow (4M)$$

Now, Thevenin's equivalent circuit seen from the terminal of the capacitor,

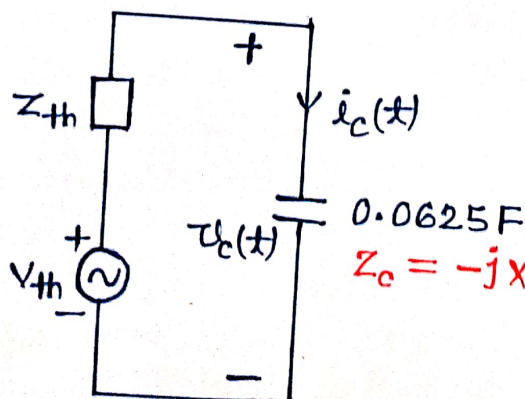


where,

$$Z_{th} = 1.02 \angle 6.91^\circ \Omega$$

$$V_{th} = 6.14 \sin(t - 53.09^\circ) \text{ V}$$

Q1(b)÷



$$Z_c = -jX_c = \frac{-j}{\omega C} = \frac{-j}{0.0625} = -j16 \Omega$$

$$\therefore v_c(t) = \left(\frac{Z_c}{Z_c + Z_{th}} \right) V_{th}$$

$$= \left(\frac{-j16}{-j16 + 1.02 + j0.12} \right) 6.14 \angle (-53.09^\circ)$$

$$= (1.00 - j0.06) \times 6.14 \angle (-53.09^\circ)$$

$$= 1.00 \angle (-3.68^\circ) \times 6.14 \angle (-53.09^\circ)$$

$$= (3.38 - j5.16) \text{ V}$$

$$= 6.17 \angle (-56.76^\circ) \text{ V}$$

$$= 6.17 \sin(t - 56.76^\circ) \text{ V} \quad \rightarrow (1.5 \text{ M}) \quad + (1.5)$$

$$\therefore i_c(t) = \frac{v_c(t)}{Z_c}$$

$$= \frac{(3.38 - j5.16)}{(-j16)}$$

$$= (0.32 + j0.21) \text{ A}$$

$$= 0.38 \angle 33.23^\circ \text{ A}$$

$$= 0.38 \sin(t + 33.23^\circ) \text{ A} \quad \rightarrow (1.5 \text{ M}) \quad + (1.5)$$